Point-of-Care Breast Cancer-on-Chip Diagnosis Using Linear SVM Model

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***Abstract*—Breast cancer is the second-leading cause of cancer worldwide. Researchers have utilized machine learning classifiers on FPGA to increase the effectiveness of breast cancer diagnosis. The proposed approach features an optimized real-time breast cancer detection system utilizing a Support Vector Machine (SVM) linear classifier for point of care. To minimize hardware complexity, this method incorporates a wrapper-based feature selection technique, identifying the most significant features, and utilizes an approximated adder design. The system achieves an accuracy of 96% while employing a reduced feature subset. FPGA implementation demonstrates a 14% reduction in hardware requirements compared to existing methods. For a single-chip solution in 45nm technology, the system achieves an area of 63,903 µm², power consumption of 43 mW, and a delay of 6.6 ns. Overall, the proposed system integrates high accuracy with reduced hardware requirements, enabling efficient real-time diagnostics in resource-constrained environments with a target of point-of-care diagnostic tools.**

***Index Terms*—Point-of-care, Feature extraction, Support vector machines, Hardware acceleration.**

I. Introduction

About 30% of all cancer cases worldwide are breast cancer, making it the second most frequent kind of cancer. In 2022, 2.3 million women received a breast cancer diagnosis, and 670,000 people died from the deadly illness [1]. To lower the fatality rate and disease progression, early detection is critical. Timely diagnosis enables more effective treatments, significantly improving patient outcomes. Recent advancements in machine learning (ML) have developed accurate and reliable models for diagnosing and treating breast cancer at earlier stages. Deep learning has found significant application in breast cancer screening by analyzing images from X-rays, ultrasounds, MRIs, and biopsies. However, deep learning models typically require large, labeled datasets and substantial computing power, which can be a limitation for many medical applications. For smaller or medium-sized datasets, Support Vector Machines (SVM) have emerged as a widely used technique, offering strong performance for breast cancer diagnosis in resource-constrained settings.

SVMs are a powerful supervised learning technique widely used for classification and regression tasks [2]. With the rise of edge computing, hardware-based SVM models enables on-device processing of data without needing to rely on servers. Implementing SVM in hardware is essential for enhancing performance, efficiency, and scalability in systems that require real-time, low-power machine learning solutions, especially in embedded systems. Lopes *et al.* [3] suggested a hardware/software co-design to implement the SVM classification system onto FPGA with the goal to detect melanoma on a chip. Yacoub *et al.* [4], suggest a reconfigurable SVM hardware architecture for varied dataset sizes in kernel and linear scenarios. In order to identify and categorize, the researchers have suggested a computer model for breast cancer or CAD (computer-based diagnostic) system. [5]. In terms of hardware, a diagnostic technique based on digital mammogram diagnostic convolutional neural network (DMD-CNN), has been suggested for the segmentation of the breast mammography dataset [6]. The PYNQ board based implementation optimizes resource utilization as well as power consumption. For hardware-accelerated breast cancer detection, FPGA-based designs have been proposed utilizing a reconfigurable neural network with higher computation complexity [7]. Moreover, a SVM linear classifier is proposed for FPGA implementation of breast cancer detection for medium size database [8] which considers entire feature values for diagnosis. An FPGA integrated with deep learning algorithms has been implemented for real-time breast cancer classification, addressing the increasing model complexity associated with deep learning techniques [9]. Implementing breast cancer classification on a point-of care chip offers significant advantages, including real-time processing, portability, and energy efficiency, making it ideal for point-of-care diagnostics. However, a combining software optimization along with hardware can further enhance the effectiveness and improve the overall efficiency of breast cancer diagnostic systems.

The proposed approach employs SVM linear classifier for breast cancer detection building upon the research outlined in the article [8]. Linear SVMs are simpler and faster for training and requires fewer hyperparameters, and suitable for real-time or resource-constrained environments. Additionally, a feature selection method and approximate hardware design are incorporated to further reduce computational complexity. The hardware implementation demonstrates superior performance compared to existing approach, enhancing both efficiency and accuracy in breast cancer diagnosis. Additionally, an Application-Specific Integrated Circuit (ASIC) design is introduced, targeting a single-chip solution, which minimizes the resource area, and provides the way for compact, high-performance, and low-power diagnostic systems targeting real-world medical applications. The goal is to extend diagnostic capabilities to settings where mammography is not immediately available. The ASIC implementation enables the system ideal for point-of-care deployment even in handheld scanner that works with a cytology probe.

Section II outlines methodology employed in the study. Section III presents the results and offers a comprehensive discussion, while Section IV provides the concluding remarks.

II. Methodology

Figure 1 illustrates the block diagram of the proposed method, where the collected breast cancer data is processed by an on-chip module. This module efficiently analyzes the data to classify subjects as either benign or malignant, enabling real-time diagnosis with optimized hardware implementation.

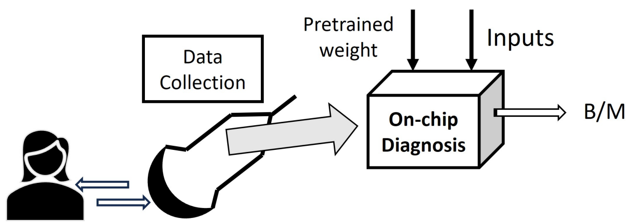


Fig. 1: Block Diagram for proposed method

We have utilized the Breast Cancer Wisconsin (Diagnostic) data set [10], which includes 569 samples with 30 features representing characteristics of cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, and concavity. The pre-processed dataset contains 357 Benign (B) and 212 Malignant (M). Features are extracted from an image obtained from a fine needle aspirate (FNA) of breast mass. In point-of-care form, the dedicated chip can provide real-time results right after an FNA sample is analyzed with a portable cytology tool. The cytology probe captures cellular information using miniature camera, and the acquired signals are passed through a postprocessing comprising analog front-end, amplification, filtering to produce digital data. These digitized feature values are transferred to the hardware module for final on-chip diagnosis. Based on the feature values, a linear SVM is used to classify if breast cancer category as benign or malignant. Finding a hyperplane in a dataset that maximum divides two classes is the goal of linear SVM. For classification, SVM makes predictions based on the sign of the decision function.

Let $\mathbf{w}$ weight vector, $\mathbf{x}$ feature vector, and b bias term. The function $f(x)$ can be represented as:

$$f(x) = \mathbf{w}^T\mathbf{x} + b$$

If $f(x) > 0$, the data point belongs to one class. If $f(x) < 0$, it belongs to the other class.

The regularization parameter (C) of the model is set at value 1. In addition, feature selection process is computed to select important features for improving model performance and reducing complexity. A Recursive Feature Elimination (RFE), a wrapper method, is utilized to iteratively select the most important features, eliminating redundant variables. The wrapper-based approach reduces the original 30 features to 25. The SVM model is trained using the selected features to classify tumors as benign or malignant. Support vectors, which define the boundary between classes, are reduced during the feature selection process, making the model more efficient. We have found 29 and 24 support vectors for 30 and 25 number of feature subsets.

For hardware implementation, the features are defined using IEEE 754 single-precision floating-point format, which ensures accurate representation of the values. The hardware design incorporates 32-bit signed floating-point multipliers and adders with a 23-bit mantissa, an 8-bit exponent, and 1 sign bit for precision. In multiplication, we put 1 extra bit in front of mantissa to become 24 bit value. Multiplication of two 24 bit mantissa used unsigned 24×24 bit multipliers to produce a result of 48 bit. For the exponent part, two exponents are added and subtracted from the bias value of 127 using 8-bit unsigned adder and subtractor. The leading one at bit 46 normalises the number and no shift is needed. For leading one at bit 47, the intermediate product is shifted to the right and the exponent is incremented by 1. So, the normalizer result becomes 30-bit and the 32 bit floating point multiplication result is found with sign bit. In addition, the reduced feature set and number of support vectors optimize memory usage and computational efficiency, resulting in a robust system for diagnosis. The proposed method is the suggested approach for classifying the WBCD dataset as benign or malignant, as shown in Fig. 2:

1) Analyze the WBCD dataset for processing. The patient data and weight values are loaded into two files: data (N) and coefficients (M) and 'n' is number of features.
2) To represent floating points, convert the real values into IEEE 754 single precision format.
3) Design of a ROM controller, multiplier, and floating point adder.
4) Linear SVM architecture is implemented using an adder tree.
5) Finally, the bias value is added to the output (Y) for the classification of benign and malignant.

A. Approximate Adder

The SVM design employs half adders and subtractors as fundamental building blocks for full adders and subtractors, which are subsequently utilized in ripple carry adders and subtractors. These modules are further integrated into floating-point multipliers and adders, incorporating a normalizer. For better hardware realization, the Approximate Adder (AA) design is also employed in the design [11]. AA's Sum is exact for 6 out of 8 possible input combinations, while the Carry-out (Cout) is accurate for all cases. The underlying logic equations governing AA's operation are:

Sum = Cout' and Cout = AB + BCin + ACin

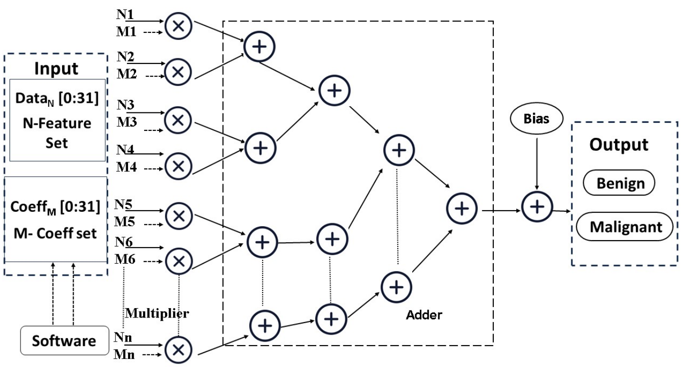


Fig. 2: SVM Flow Diagram for Classification

III. RESULTS

The SVM algorithm is employed to develop a breast cancer detection system, evaluated using two distinct feature sets: one comprising all 30 available features and another with a reduced set of 25 features. Dataset is divided as 70% training as well as 30% testing. The performance metrics as shown in Table VI shows that accuracy (ACC), sensitivity (SEN), and specificity (SP) of the reduced-feature SVM model is found comparable with that of all features.

TABLE I: Performance Metrics for Breast Cancer Detection Using SVM

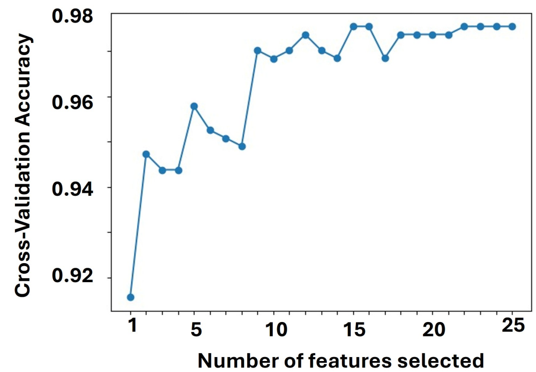
Feature Set	ACC	SEN	SP	Computational
All Features (30) [8]	0.95	0.98	0.90	31M + 30ADD
Reduced Features (25)	0.96	0.98	0.92	25M + 24ADD

TABLE II: 10-fold cross-validation Performance of Different Classifiers

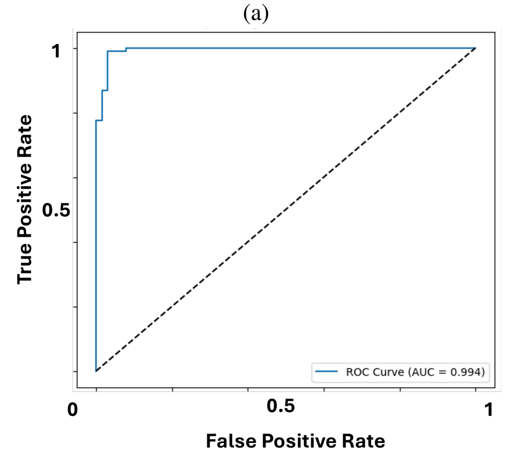
Model	ACC	SEN	SP
SVM (Linear)	0.98	0.99	0.96
Decision Tree	0.93	0.94	0.91
Neural Network (MLP)	0.98	0.97	0.98

Moreover, the results shown in Table I demonstrates that reducing the feature set leads to significant reductions in number of multiplier and adder. A 10-fold cross-validation analysis using the reduced feature subset, along with multiple low-complex machine-learning classifiers (Like: Decision tree and Neural network), is summarized in Table II. The Recursive Feature Elimination (RFE) wrapper method was employed to rank feature importance, and the resulting accuracy trend for different feature subsets is shown in Fig. 3(a), where the model achieves its better performance towards 25 features. Additionally, the robustness of the proposed classifier is confirmed through the Receiver Operating Characteristic (ROC) curve illustrated in Fig. 3(b), which achieves an Area Under the Curve (AUC) of 0.99.

The FPGA based implementation is computed in Xilinx CPG236 Artix-7 FPGA board and hardware utilization report is represented in Table III. Additionally, the utilization



(a)



(b)

Fig. 3: Model performance (a) Accuracy based on feature subset, (b) AUC-ROC curve for overall performance

report for the approximate adder design highlights a slight reduction in the number of LUTs compared to the standard design, further optimizing hardware resources while maintaining performance. This balance between feature reduction and approximate hardware design contributes to an efficient implementation.

TABLE III: Hardware and Approximate Hardware Resource Utilization

Features	Slice LUTs	Slice Registers
25 Features (Actual)	14,263 (68.57%)	832 (2.00%)
25 Features (Approx.)	14,244 (68.48%)	832 (2.00%)
30 Features (Actual) [8]	17,201 (82.70%)	992 (2.38%)
30 Features (Approx.) [8]	17,196 (82.67%)	992 (2.38%)

The software output obtained from the SVM linear classifier is shown in Table IV and Table V for 30 and 25 features respectively. The generated 32-bit hexadecimal number is translated into a real number for comparison, and the hardware and software findings nearly match for an accurate diagnosis. The result shows just six randomly selected samples for testing analysis and comparison with the WBCD labelled dataset. However, the approximate based design sometimes overesti-

TABLE IV: Output of SVM linear classifier algorithm using 30 features

Subjects	$Y_{Software}$	$Y_{Hardware}$	$Y_{Approx\ Hardware}$	Diagnosis
Sub1	-4.5294	-4.5292	-6.2179	B
Sub2	0.3574	0.35773	0.54178	M
Sub3	-0.6159	-0.6165	-1.0206	M
Sub4	-0.3477	-0.3473	-4.8375	M
Sub5	-0.7511	-0.7508	-1.7755	M
Sub6	-4.3058	-4.3056	-11.3623	B

TABLE V: Output of SVM linear classifier algorithm using 25 features

Subjects	$Y_{Software}$	$Y_{Hardware}$	$Y_{Approx\ Hardware}$	Diagnosis
Sub11	5.3181	5.43555	3.54355	M
Sub12	-10.053	-9.9556	-9.9556	B
Sub13	6.02648	6.11982	5.67795	M
Sub14	-4.5046	-4.4004	-4.2927	B
Sub15	-4.6423	-4.5303	-3.6013	B
Sub16	3.0107	3.1320	**-1.0635**	M*

mates or underestimates which has been reflected for Subject 16 in Table V. The approximate based design wrong classifies the subject. Based on the analysis, while the approximate adder offers a slight reduction in hardware complexity compared to other methods, it also leads to misclassification in the diagnosis. Therefore, utilizing a reduced feature subset proves to be a more effective and reliable approach for diagnosis, surpassing both the approximate hardware method and the full dataset.

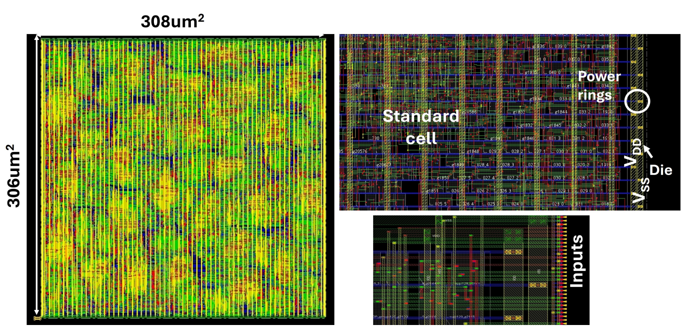


Fig. 4: Detailed layout of breast cancer on chip

TABLE VI: Methods in 45nm technology process

Feature Set	Delay(ns)	Power(mw)	Area(μm²)
All Features (30) [8]	6.8	51	78626
Reduced Features (25)	6.6	43	63903

Furthermore, the architecture is synthesized using the Cadence 45nm technology @142MHz process to optimize performance and resource utilization. The layout of the reduced-feature-based model is depicted in figure 4, showcasing the integration of the proposed design. Table VI provides a detailed comparison of key performance parameters, including delay, power consumption, and chip area, highlighting the efficiency of the ASIC implementation. The design was synthesized using Cadence Genus and implemented using Cadence Innovus 22.1 with the 45-nm standard-cell library. Reported power values are now explicitly stated as post-layout estimates based on gate-level simulation activity. The power was estimated using Cadence Innovus signoff power analysis. Switching activity was generated from gate-level simulations (VCD/SAIF) using Export synthesized netlist and SDF timing file from Genus. The tool computes internal power, switching power, and leakage power using the post-layout netlist, extracted parasitics (SPEF), and real activity data.

Overall, this research successfully achieves its goals of designing an efficient, hardware-friendly model for breast cancer detection targeting point-of-care diagnostic tools. The findings suggest that feature selection techniques are crucial in balancing performance with resource efficiency, particularly for deployment in embedded systems. However, the current reduced feature set still comprises 25 input parameters for the SVM model, contributing to higher power consumption. Future work could focus on refining feature selection techniques to reduce dimensionality further while maintaining diagnostic accuracy. Additionally, optimizing hardware approximation techniques, particularly for multipliers, could enhance energy efficiency and computational performance, addressing the growing need for low-power, high-speed medical diagnostic tools. For point of care on chip diagnosis, the features values can be extracted from the sensor dataset and on-chip tool provides diagnosis for malignant and benign classes.

IV. Conclusion

The paper presents the implementation of the linear SVM algorithm for detection of breast cancer as Benign and Malignant. The implementation computed in different ways to reduce computation complexity as well as hardware utilization. The feature selection method has been applied to select the important features for classification and result produces the better performance compared to the literature. An approximate adder is also designed to reduce the computation complexity. However, its has been found that inclusion of approximate adder reduces the hardware slightly but it can increase the chances of misclassification. In future, we will include optimized multipliers as well as adder to achieve hardware as well as energy efficiency, with the guarantee of consistent performance on real world dataset. Overall, the proposed system based on reduced feature subset achieves a balance between performance and efficiency, demonstrating the potential for enhanced breast cancer detection in resource-constrained environments. However, the study is focusing on an embedded diagnostic chip in a portable cytology system to enhance early detection efforts, especially where breast cancer is diagnosed late due to screening unavailability.

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